

Prioritizing Search Content Decay: A Leak-Free Learning-to-Rank Engine

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Abstract

How can enterprise search editorial teams with strict review capacities (e.g., 50 pages per month) prioritize which published assets to update among tens of thousands of degrading URLs? Using an anonymized multi-domain dataset of 30,000 published pages across 200 client websites from the FlyRank Search Intelligence Warehouse, we formulate content refresh prioritization as a Learning-to-Rank task evaluated by Precision@50 under client-holdout validation. While standard industry heuristics (refreshing stale pages with high historical traffic) achieve a Precision@50 of only 0.240—underperforming the 0.542 population base rate due to survivorship bias on mature evergreen assets—our leak-free Random Forest ranking engine achieves a Precision@50 of 0.680 (a **2.83x lift**, yielding +22 additional correctly prioritized declining assets per cycle). Target leakage is strictly prevented by excluding retrospective slope indicators (trend_pct, trend_direction), and cross-domain memorization is eliminated through an 80/20 grouped client split (27,675 train rows across 160 clients, 2,325 holdout rows across 40 unseen clients). These rankings directly drive an explainable editorial review queue categorized by diagnostic reason codes, providing decision-support intelligence without unsubstantiated causal claims.

1. Problem Framing & Asymmetric Capacity Constraints

Enterprise search domains hold 10,000 to 50,000 published pages, yet editorial teams can thoroughly audit and rewrite only 20 to 50 assets per month. Manual heuristics (e.g., sorting by age or traffic drop) suffer severe error costs: false positives waste 8 hours of copywriter labor (\$400–\$800) on stable assets, while false negatives allow high-volume commercial pillars to surrender Page 1 SERP rankings to competitors. Our unit of analysis is a single published asset (content_id) within a client domain (client_id).

2. Data Safety & Leakage Guardrails (FlyRank Warehouse 30k Cohort)

Field Category	Representative Columns	Operational Protocol & Rationale
Pre-Decision Features	content_age_days, days_since_last_update, log_impressions_90d, ctr, avg_position, word_count	Retained for model training. Strictly observable prior to editorial review timestamp.
Pseudonymous IDs	client_id, content_id	Excluded from feature sets. Used exclusively for grouped 80/20 client-holdout splits.
Target-Derived Slope	trend_direction, trend_pct	Strictly banned from features. Retrospective slope introduces fatal leakage (false 1.000 score).
Customer PII	URLs, client names, raw search queries	Zero PII present. All entities hashed into pseudonymous identifiers.

3. Client-Holdout Benchmark Performance (K = 50 Editorial Budget)

Evaluation Metric	Population Base Rate	Hand-Rule Baseline	Decision Tree	Random Forest (Best)	Empirical Lift
Precision@20	0.542 (54.2%)	0.150 (3 / 20)	0.500 (10 / 20)	0.700 (14 / 20)	4.67x
Precision@50 (Target)	0.542 (54.2%)	0.240 (12 / 50)	0.540 (27 / 50)	0.680 (34 / 50)	2.83x
Precision@100	0.542 (54.2%)	0.360 (36 / 100)	0.530 (53 / 100)	0.700 (70 / 100)	1.94x
Holdout ROC-AUC	0.500	0.627	0.742	0.747	+0.120
Average Precision	0.542	0.468	0.575	0.610	+0.142

4. Feature Importance & Interpretation

Evaluation across 52 pre-decision features reveals that search presence consistency (days_with_impressions: 16.0%) and visibility scale (log_impressions_90d: 12.8%) dominate decay probability. Average SERP position drift into striking distance (avg_position: 10.8%) and content age (content_age_days: 9.5%) provide contextual nuance. Crucially, content age alone shows near-zero linear correlation (+0.04) with decay, explaining why age-only heuristics consistently fail.

5. Action Playbook: Diagnostic Reason Codes & Queue

Diagnostic Reason Code	Trigger Condition (Observable Signals)	Recommended Editorial Intervention
stale_visible_page	Impressions >5,000 and Days Since Update >180	Comprehensive Refresh: Fact-check statistics, update examples, and re-index.
page_one_decay_risk	Average Position 1–10 with softening engagement	Defensive Refresh: Reinforce internal backlinks and align search intent.
thin_visible_page	High impression volume with Word Count <1,200	Content Expansion: Deepen topical coverage and add technical diagrams.
low_ctr_visible_page	High impressions but CTR <0.5%	Metadata Optimization: Rewrite SERP title tag and compelling meta description.

6. Limitations & Honest Framing

Predictive, Not Causal: The model identifies observational patterns associated with historical decay. It does not prove that rewriting an asset will automatically restore traffic. Validating refresh ROI requires randomized controlled trials (A/B testing).

No Algorithm Reverse-Engineering: The system operates strictly as decision-support capacity allocation for human editors.

Unobserved SERP Shifts: Page-level metrics do not record external Google AI Overviews, rich snippets, or broad core algorithm updates.

7. Reproducibility & Audit Trail

All results are reproducible from fresh clones using Python 3.11/3.12 and random_state=42:

```
git clone https://github.com/harshitttt077/flyrank-ml-internship-starter.git && python scripts/run_all.py
```

- **Committed Metrics Receipt:** outputs/model_results.json
- **Executed Capstone Notebook:** work/notebooks/capstone.ipynb (all cells executed top-to-bottom)
- **Full Markdown Report:** work/capstone_report.md | **Deployed Paper URL:** submission/paper_url.txt

8. Acknowledgments & Data Credit

Built on the **FlyRank ML Internship dataset** linking to <https://flyrank.ai>. Sincere gratitude to the FlyRank engineering and search research team for providing realistic, anonymized enterprise search intelligence data.